

STAT 4710/5710: Homework 3

Ying Jin

Due: October 23, 2025 at 11:59am

Contents

Instructions	1
1 Case study: logistic regression for home loan (45 points)	3
1.1 Data import (4 points)	3
1.2 Data exploration and visualization (4 points)	3
1.3 Univariate logistic regression (23 points)	3
1.3.1 Understanding logistic regression (13 points)	3
1.3.2 Univariate logistic regression on the full data (10 points)	4
1.4 Multivariate logistic regression (14 points)	4
2 Case study: linear regression for housing prices (55 points)	5
2.1 Exploratory data analysis (10 points)	5
2.2 Predictive modeling preparation (1 point)	5
2.3 Ordinary least squares (6 points)	5
2.4 Ridge regression (13 points)	5
2.5 Lasso regression (10 points)	6
2.6 Evaluation and comparison (15 points)	6

Instructions

Materials and collaboration

The policy on allowed materials and collaboration is as stated on the Syllabus:

“Students are permitted to work together on homework assignments, but must write up and submit solutions individually. In particular, students may not copy each others’ solutions. Students may consult all course materials, textbooks, the internet, or AI tools (e.g. ChatGPT) to complete their homework. Students may not use solutions to problems that may be available online and/or from past iterations of the course. For each homework, students must disclose all classmates with whom they collaborated, which AI tools they used, and how they used them. Failure to do so will result in a 10-point penalty.”

In accordance with this policy,

Please disclose all classmates with whom you collaborated:

Please disclose which AI tools you used, and how you used them:

Failure to answer the above questions will result in a 10-point penalty.

Writeup

Use `homework-3-template.Rmd` as a starting point for your writeup, adding your solutions after `***Solution***`. Add your R code using code chunks and add your text answers using `**bold text**`. Consult the [preparing reports guide](#) for guidance on compilation, creation of figures and tables, and presentation quality. If the instructions ask you to “print a tibble”, you should print the tibble directly.

Programming

We encourage you to follow the `tidyverse` paradigm for data visualization, manipulation, and wrangling, as in the first homework, to practice tidyverse workflows (e.g., `filter`, `mutate`, `group_by`, `summarize`, `join`, `ggplot2`).

Students who strongly prefer to use Python for homework may do so. If you choose this option, you are **required** to present both code chunks and the corresponding outputs/interpretations in your PDF. In addition, it’s recommended that you:

- Use `pandas` for data manipulation (preferred, since it parallels `dplyr`).
- Use `plotnine` for visualization (preferred, since it parallels `ggplot2`). If using `seaborn/matplotlib`, you must still include the required plot elements (labels, 45° line, smoothing curve, facets, etc.).

```
library(tidyverse)
library(splines)
library(glmnetUtils)
```

Code completion. In some problems, the task is to complete an incomplete code chunk provided. In those problems, in the template, there is `eval = FALSE` in the chunk header to prevent evaluation errors. You should remove `eval = FALSE` after you have completed the code to enable evaluation and printing of results.

Grading

The point value for each problem sub-part is indicated. There are 100 points possible on this homework, evaluated based on the correctness of the numerical results and plots. When a problem asks for a plot:

- **Axes and labels:** Always include informative axis labels and, if appropriate, titles.
- **Required elements:** If the instructions ask for a 45° line, smoothing curve, or facets, these must be included.
- **Clarity:** Points may be deducted if plots are cluttered, unreadable, or missing labels.

Submission

Compile your writeup to PDF and submit to Gradescope.

1 Case study: logistic regression for home loan (45 points)

In this case study, we will explore logistic regression with the home equity loan (HMEQ) dataset. In the HMEQ dataset, each row is a home-equity loan applicant and the variables include:

Variable	Meaning
STATUS	Outcome status: bad = default/serious delinquency, good = paid.
LOAN	Amount of the loan request (dollars).
MORTDUE	Amount still owed on the existing mortgage (dollars).
VALUE	Appraised value of the property (dollars).
REASON	Purpose of the loan: DebtCon (debt consolidation) or HomeImp (home improvement).
YOJ	Years at current job.
DEROG	Number of major derogatory credit reports.
DELINQ	Number of delinquent credit lines.
CLAGE	Age of the oldest credit line (months).
NINQ	Number of recent credit inquiries.
CLNO	Total number of credit lines.
DEBTINC	Debt-to-income ratio.

Our goal is to predict whether the loan will go bad (default or serious delinquency) using information available at application time, so one can set terms appropriately or decline risky applications.

1.1 Data import (4 points)

1. (2 points) Import the dataset `home_loan.csv` into a tibble called `hmeq`, and drop all rows containing missing values (NAs). Print out the tibble.
2. (2 points) Redefine the outcome variable `STATUS` that is 1 if the original variable `STATUS` is bad and 0 if `STATUS` is good. After this, use the command `hmeq$STATUS = factor(hmeq$STATUS, levels = c(0, 1))` to convert `STATUS` to a factor variable. Print out the tibble.

1.2 Data exploration and visualization (4 points)

1. (2 points) To check how the debt-to-income ratio look like, make a histogram of `DEBTINC`, and add a vertical red line for the median value. Do you notice some pattern?
2. (2 points) Use a plot to visualize the distribution of `DEBTINC` for good/bad `STATUS` groups. What patterns do you notice?

1.3 Univariate logistic regression (23 points)

We'll use `DEBTINC` as the single predictor to build a logistic regression model.

1.3.1 Understanding logistic regression (13 points)

1. (1 points) Split `hmeq` into a training (80%) and a test (20%) set, using the rows in `train_samples` as the training set, and the remaining as the test set. Store them in tibbles called `hmeq_train` and `hmeq_test`. Set the random seed as in the provided code for reproducibility.

```
set.seed(4710) # seed set for reproducibility (DO NOT CHANGE)
n <- nrow(hmeq)
train_samples <- sample(1:n, round(0.8 * n))
```

2. (2 points) Define and print a new tibble called `hmeq_train_subset` containing two columns `STATUS` and `DEBTINC` for all data in `hmeq_train` whose `VALUE` value is strictly smaller than 50000 and `YOJ` value

equal to 0, and REASON is equal to 'DebtCon', and MORTDUE is strictly larger than 23000. Print out this tibble.

- (3 points) Write down the mathematical formula for the logistic regression likelihood function $\mathcal{L}(\beta_0, \beta_1)$ using the values of observations in `hmeq_train_subset`, where β_0 is the coefficient for the intercept, and β_1 is the coefficient for DEBTINC.
- (5 points) Find the MLE based on this subset using `glm()`. Given a value of DEBTINC, what is the estimated probability of $\mathbb{P}[\text{STATUS} = 1 \mid \text{DEBTINC} = x]$ based on the fitted model? Write down the mathematical formula. [Hint: after fitting the model, use `summary()` to display the coefficients and obtain the formula.]
- (2 points) How are the fitted values of β_0 and β_1 above obtained from the formula in part 2?

1.3.2 Univariate logistic regression on the full data (10 points)

- (3 points) Run a univariate logistic regression using DEBTINC as the only predictor and `hmeq_train` as the training set. Print out the summary of the fitted model.
- (3 points) Based on the fitted model above, how do the odds of default/serious delinquency change when DEBTINC increase by 1?
- (4 points) Plot the logistic regression fit along with a scatter plot of the training data. Use `geom_jitter()` instead of `geom_point()` to better visualize the data. Based on the plot, roughly what is the estimated probability of default for someone with DEBTINC = 50? [Hint: for the y value in `aes`, use `y=1*(STATUS==1)` to convert the factor variable STATUS to numeric 0-1 variable.]

1.4 Multivariate logistic regression (14 points)

- (5 points) Run a logistic regression of STATUS on all the other variables in the training dataset. Other things being equal, does the estimated coefficient suggest that applicants with higher DEBTINC are more likely to default? Other things being equal, what impact does an increase in NINQ have on the odds of default/serious delinquency?
- (3 points) For a new applicant with LOAN=10000, MORTDUE=65000, VALUE=90000, YOJ=5, DEROG=0, DELINQ=0, CLAGE=120, NINQ=1, CLNO=15, DEBTINC=50, REASON='DebtCon', JOB='Sales', report the predicted probability of default and explain it in one clear sentence (e.g., "roughly a XX% chance of default given these features").
- (6 points) To examine how well this model performs on the test data, make the predictions and threshold the estimated probability at 0.5 to predict the label for the test data, and report misclassification rate, precision, recall, and F1 score by completing the following code chunk.

```
... # TO COMPLETE
misclass_rate <- # TO COMPLETE
precision <- # TO COMPLETE
recall <- # TO COMPLETE
F1 <- # TO COMPLETE

test_errors <- tribble(
  ~Metric, ~Value,
  "Misclassification_rate", misclass_rate,
  "Precision", precision,
  "Recall", recall,
  "F1_score", F1
)
# print the tibble
test_errors
```

2 Case study: linear regression for housing prices (55 points)

In this case study, we will build linear regression models with the Ames Housing data. This dataset describes residential property sales in Ames, Iowa. It has 2,930 sales and a rich set of predictors spanning dozens of numeric and categorical features. In the dataset, the categorical features have been translated to one-hot encoding convenient for modeling.

2.1 Exploratory data analysis (10 points)

1. (1 points) Import the dataset `ames_housing.csv` into a tibble called `ames_data`. The log of `Sale_Price` is the response we are to predict. Print out the tibble.
2. (3 points) Create a histogram of `log_Sale_Price` using `binwidth=0.1`, and add a vertical red line for the median value. Describe the distribution of the log sale price.
3. (6 points) Make two separate scatter plots (using `alpha=0.2` for better visualization) of `log_Sale_Price` against: `Gr_Liv_Area` (above-ground living area) and `Year_Built`, and add a smoother. For each plot, write a sentence describing the pattern.

2.2 Predictive modeling preparation (1 point)

Now we will build some predictive models for `log_Sale_Price`.

To prepare, split the data into a training (80%) and a test (20%) set, using the rows in `train_samples` as the training set, and the remaining as the test set. Store them in tibbles called `ames_train` and `ames_test`. Set the random seed as in the provided code for reproducibility.

```
set.seed(4710) # seed set for reproducibility (DO NOT CHANGE)
n <- nrow(ames_data)
train_samples <- sample(1:n, round(0.8 * n))
```

2.3 Ordinary least squares (6 points)

1. (2 points) Using the training set `ames_train`, run a linear regression of `log_Sale_Price` on all the other features in the dataset. Display the summary of the fitted model (this will print a long summary).
2. (2 points) Why are there NA values in the fitted coefficients?
3. (2 points) Do the sign of the estimated coefficients for `Gr_Liv_Area` and `Year_Built` match the patterns we see in the previous two plots? Explain your findings.

2.4 Ridge regression (13 points)

1. (4 points) Using the training set `ames_train`, run a ridge regression of `log_Sale_Price` on all the other features in the dataset with 10-fold cross-validation. Plot the cross-validated mean squared error as a function of λ . What is the value of lambda selected according to the one-standard-error rule?
2. (2 points) During the above cross-validation process (excluding any subsequent refitting to the whole training data), how many ridge regression models were fit? For the purposes of this question, a ridge regression fit is the process of solving the ridge optimization problem for a single value of λ . [Hint: in the `cv.glmnet()` function, the default number of candidate λ values is 100.]
3. (2 points) Following on part 2, for every observation in the dataset, during the above cross-validation process (excluding any subsequent refitting to the whole training data), how many ridge regression models were fit using it?
4. (5 points) Use `plot.glmnet()` function provided here to visualize the the ridge regression fitted coefficients, highlighting 6 features using the `features_to_plot` argument. Do any of those 6 features change sign as λ decreases? Do they change monotonically as λ decreases?

2.5 Lasso regression (10 points)

1. (3 points) Fit a 10-fold cross-validated Lasso regression to the training data and display the CV plot.
2. (3 points) What is the value of λ selected according to the one-standard-error rule? How many features have nonzero coefficients in the Lasso fit with this value of λ ?
3. (4 points) Use `plot_glmnet()` to visualize the Lasso fitted coefficients, and set `features_to_plot=8` to highlight 8 features selected by the Lasso. By examining this plot, answer the following questions. Which feature is the first to enter the model as λ decreases? Which feature has the largest absolute coefficient for the most flexibly fitted Lasso model?

2.6 Evaluation and comparison (15 points)

1. (4 points) Evaluate the OLS, ridge, and Lasso models on the test set `ames_test` using mean squared error (MSE). Print the results using the following incomplete code. Which model performs best on the test data?
2. (4 points) Based on the above result, as well as the CV curves for ridge and Lasso, do we suspect that the bias or variance is the dominant force in driving the test error? Why? Does this make sense, given the number of features relative to the sample size?
3. (4 points) Collect the least squares, ridge, and Lasso coefficients, excluding the intercept, into a tibble called `coeffs` with columns `feature`, `ls_coef`, `ridge_coef`, `lasso_coef` (using `s = 'lambda.1se'` for ridge and Lasso). Sort the rows based on the absolute value of `ls_coef` (in the descending order). What patterns do you notice when comparing the magnitudes of coefficients from OLS and Ridge?
4. (3 points) Print the subset of rows in `coeffs` where the OLS fitted coefficient is NA. Do these features still have NA values in their fitted ridge and Lasso coefficient? How did ridge and Lasso address the issue that leads to NAs in OLS?